The University of Hong Kong

FITE4801 Final Year Project

A Decentralized Data Wallet Unlocking Individual Data into the Next Digital Asset

Detailed Project Plan

Luk Wing Kei (3033104368)

Wong Tsz Tung (3036065408)

Table of Contents

1. Introduction

Motivated by the escalating recognition of personal data as a valuable asset in our contemporary data-driven economy, this project explores the design and development of a decentralized data wallet (DDW) to empower individuals with greater control and ownership over their data, enabling transparent peer-to-peer (P2P) data sharing with enhanced privacy controls. This further unlocks data monetization opportunities while simultaneously providing enterprises with reliable and ethically sourced insights.

1.1 Project Overview

The concept of personal data sovereignty has emerged as a global focal point in the era of widespread digitalization, fueled by increased data collection and the rising demand of tech giants and centralized platforms for generating commercial insights, targeted advertising, and profit extraction from user data [1], [2]. Personal data has become one of the most valuable digital assets in today's digital economy, which is estimated to generate trillions in revenue for tech giants annually [3]. This is further reaffirmed by the World Economic Forum [4], which stated that personal data has emerged as a new asset class, creating unprecedented economic prospects, and making individuals eligible to capitalize and monetize their personal data.

Acquisti, Taylor, and Wagman [5] reveal that user data analytics and behavioral tracking through predictive modelling have become central to business models that maximize advertising efficiency and long-term revenue growth. Together, these assert the view that personal data functions as both a strategic corporate resource and a potentially monetizable personal asset from both an individual and corporate perspective.

1.2 Project Motivation

Despite the immense economic significance of personal data, individuals, as data owners, rarely retain meaningful ownership, control, or financial benefit from their personal data. Pew Research Centre in the United States [6] revealed that 81% of Americans have expressed that they lack personal control over the data collected by companies, and over half of them have little or no understanding of which companies and how they are going to use and deal with their collected personal data. This indicates data owners are left alone in their control on how their data are used and handled by the companies, to whom they are shared with, for what purposes their data are collected, and the extent to which enterprises, as data consumers, can extract value or insights from their personal data and in what ways that relevant insights could otherwise help companies generate additional economic value.

Meanwhile, centralized platforms and enterprises continue to dominate the data ecosystem. They collect data from users, extracting disproportionate profits from their personal information without providing fair compensation or transparency of the usage of data to the original data owners [7]. This dynamic reinforces data silos and perpetuates information asymmetries, and further contributes to unequal monetary value distribution between individuals and enterprises as data consumers in the emerging digital economy [8].

Existing approaches, such as data aggregators and blockchain-based peer-to-peer marketplaces, have advanced the broader field of data trading, but they remain limited in offering user-centric solutions. Javelin Strategy & Research [9] acknowledges the potential of tokenizing identity-related data through blockchain to create a secure and user-controlled identity framework to address problems in identity management, such as a lack of user control over how identity information is shared or verified, and fragmented and duplicated identity records. Yet, it does not provide any mechanisms for users to monetize data. Similarly, Ocean Protocol demonstrates the feasibility of tokenized data assets by providing infrastructure for data exchange, but its architecture does not address the problems of personal data sovereignty [10]. McKinsey's analyses further underline the economic promise of tokenization, but it also identifies gaps in designing comprehensive architectures to balance privacy, user empowerment, and enterprise needs for actionable analytics [11].

Building on these insights, our final-year project draws inspiration from DataMesh+ suggested by Merlec and In [12], a blockchain-powered peer-to-peer exchange model for self-sovereign data marketplaces, while extending the vision through a decentralized data wallet. Unlike marketplace-focused models, our decentralised data wallet empowers individuals to tokenise and monetize their personal data under secure, consent-driven management while enabling valuable insights to enterprises for industry applications. Thus, this enables a mutually beneficial data ecosystem where individuals gain control and economic value from their personal data, while enterprises access actionable insights for improved decision-making and innovation. This unresolved gap forms the central motivation for our project.

1.3 Project Objective

The project responds to a systemic challenge in the data ecosystem, where individuals generate vast amounts of personal data yet receive no fair share of its value, while enterprises continue to face persistent issues of data quality, trust, and consent compliance. By developing a blockchain-powered decentralized data wallet, we aim to implement this new conceptual data architecture that reconfigures data ownership, sharing, and monetization mechanisms.

The core objective of this project is to conceptualize, design, implement, and evaluate a DDW that empowers individuals to securely upload, store, manage, and monetize their personal data while delivering anonymized actionable insights suitable for enterprises through an insight-generating model. By doing so, this allows individual to exercise greater control over their data ownership and exercise stronger authority over personal data sovereignty, and enterprises can benefit from access to anonymized, high-quality insights.

1.4 Project Key Concept Explanations

1. Decentralised Data Wallet

A decentralised data wallet is a secure and user-controlled digital tool that enables individuals to securely store, manage, and fully own their personal data in a self-sovereign, privacy-preserving manner without relying on a centralised authority [13]. Unlike traditional systems, data is stored in centralized databases controlled by corporations, governments, or service providers. A decentralized data wallet aims to address the shortcomings of the existing centralized data infrastructures, including limited user control over how their data is collected, stored, used, or shared. It also mitigates privacy risks and prevents monetization of personal information without consent.

By leveraging distributed ledger technologies (DLTs), such as blockchain, and a decentralized oracles network, users can choose to upload their personal information, selectively choose what information they want to disclose to businesses, and give access to whom or which party on a peer-to-peer network [13]. In essence, it functions like a bank account for personal data: the user holds the private keys and controls access to each portion of their data.

Key characteristics of decentralised data wallets:

- **Peer-to-Peer (P2P) Transfers**: Allows direct P2P exchange and delegated access of personal data between individuals (data contributors) and enterprises (data consumers) without the need for a central intermediary. When a data consumer initiates a request to view certain data in an individual's wallet, the individual's wallet, operating on a separate node, can selectively approve, deny, or provide limited access to the requested data.
- **User-Centric Control**: Individuals manage access via cryptographic keys, enabling selective sharing without third-party oversight and retaining exclusive rights to share their data [14].
- **Self-sovereign identity (SSI)**: Integration with decentralized identifiers (DIDs) allows individuals to authenticate themselves or share credentials without exposing unnecessary information [14], [16].

- **Selective disclosure and privacy**: Advanced cryptographic techniques, such as zero-knowledge proofs or anonymization, enable users to share only the minimum information required for a transaction, thus proving eligibility without revealing personal details [15].

The details of the technical protocol of the decentralised data wallet will be explained in a later section to illustrate how these layers interact before proceeding to system development.

2. Blockchain

Blockchain is a decentralised and distributed public ledger that securely stores transaction records across a network of nodes, making the transaction records cannot be changed or altered retroactively. Once a transaction is initiated, nodes validate and record the transaction through a consensus mechanism, allowing all nodes in the network to agree on the state of the ledger. Once the transaction record is added to the blockchain, the record will be confirmed and becomes immutable, making it permanently recorded across the distributed ledger. This creates a chain of unchangeable, transparent, and tamper-resistant records that is shared across all nodes [17]. Thus, such a mechanism ensures transparency, data integrity, and trust.

Blockchain is primarily used for Bitcoin to make peer-to-peer digital currency transfers possible and secure [17], yet its application has been extended to data transactions and data sharing, such as a decentralised P2P data marketplace and a decentralised data wallet.

Key characteristics of blockchain includes:

- **Decentralisation**: Unlike centralized databases, blockchain relies on a distributed network of nodes that collectively verify and validate transactions to enhance security and fault tolerance [18].
- **Immutability**: All the nodes keep the transaction records on the blockchain, making them cannot be modified or deleted, ensuring trust in the network.
- **Distributed Ledger**: Every participant in the network maintains a copy of the ledger, providing complete transparency and fast updates across the network when changes occur [18].

Blockchain is the cornerstone of the decentralised data wallet. Leveraging blockchain ensures data access logs and permissions are auditable, tamper-proof, and transparent to the data owner and data contributors. Enterprises, as data consumers, can grant or revoke access to data via smart contracts, which automatically execute and log consent for commercial insights. In addition, it makes self-sovereign identity (SSI) possible, allowing individuals to prove credentials or identity attributes, such as age or address, without exposing full datasets or relying on third-party intermediaries. Thus, the decentralized nature of the data wallet gives individuals greater ownership and sovereignty over their data, preventing centralized misuse and promoting trust in digital interactions.

3. Personal Data Sovereignty

Personal Data Sovereignty is defined as an individual's ability to autonomously control the access, use, sharing, and governance of their personal data assets across different technological and organizational boundaries, regardless of where the data physically resides [2]. It fundamentally empowers individuals to govern their personal information autonomously, including deciding who can access it, under what conditions, and for what purposes. The concept of personal data sovereignty extends beyond mere regulatory compliance to enable meaningful control over personal data generation, distribution, and monetization [2],[19].

The idea of personal data sovereignty seeks to resolve three major underlying shortcomings in today’s data ecosystem, which correspond and echo with the project's motivation:

1. **Data fragmentation:** Personal data is often scattered across multiple platforms and service providers, making it difficult for individuals to maintain comprehensive oversight or control over their data [19].
2. **Lack of granular consent mechanisms**: Current data systems provide limited options for individuals to specify how, when, and to what extent their data may be accessed or utilized, restricting meaningful user agency that limits individuals’ real power to decide what happens with their data [19].
3. **Economic inequity**: Individuals rarely capture any financial value from their own data, despite growing recognition of personal data as a key digital asset in the global digital economy.

By addressing these challenges, personal data sovereignty seeks to restore user autonomy, transparency, and equitable participation in the data ecosystem.

4. Data Provenance

Data provenance refers to the documented, complete historical record of a piece of data, capturing its origin, creation, movements, modifications, and transformations it has undergone throughout the data's lifecycle [20]. It answers the questions related to (1) "Where did the data come from?", (2) "Who is the creator or owner of this piece of data?", (3) "When has the data been changed?, (4) "how has the data been accessed, processed, or changed?" and (5) "who has had control or custody of it?".

The primary goal of data provenance is to ensure the authenticity, integrity, and verifiability of the data by recording a comprehensive history of its creation, changes, and the sources it was derived from [20]. It aims to provide transparency and accountability in data handling, thereby increasing trust and confidence in data used for decision-making.

It is essential to understand why data provenance is important to our project since it is one of the most fundamental problems that we would like to address in our decentralised data wallet, where enterprises risk using data with bad quality or unreliable data for decision-making [20],[21]. Only when data are voluntarily contributed and directly provided by data owners, and subsequently verified through trusted mechanisms, such as oracles or third-party validation, can the dataset be considered accurate, reliable, and fit for use [21]. Under these conditions, the derived insights hold sufficient credibility and commercial value to inform business decision-making processes.

Key characteristics of data provenance:

- **Traceability (Origin tracking)**: Tracks where the data comes from by linking it and identifying its root source to ensure data credibility and authenticity [20].
- **Accountability**: Identifies the actors (individuals, organizations, or systems) responsible for generating, modifying, or using the data, thus providing attribution and responsibility for data usage, which is crucial for trust and auditing [20].

5. Data Monetization

Data Monetization is the process of converting data into measurable economic value or revenue streams. Data monetization can be broadly divided into two categories:

- **Internal data monetization**: Extract value from data to improve the organization's internal operation efficiency and decision-making, rather than generating revenue directly from data.
- **External data monetization**: Generate revenue by providing value-added data services to external stakeholders or third parties.

Traditionally, this involved the direct sale of datasets or using data to enhance internal operational efficiency, which mainly focused on internal monetization practices. As digital economies continue to evolve, organizations have recognized that personal data, including identity, demographic profiles, online behavioural records, and transaction histories, is not merely a passive by-product. Instead, they could and should be treated as economic assets that can be appraised, managed, transferred, and monetized.

Over time, enterprises have extended focus towards external data monetization, such as offering analytics, licensing controlled access to data, embedding data-driven intelligence into products or processes, and developing new services [23],[24]. This has initially created opportunities for new market segments such as Insights-as-a-Service (IaaS), which delivers processed, value-added analytics or intelligence derived from raw data, rather than the data itself, through scalable digital platforms. This model responds to organizations' growing institutional and commercial demand for data-driven intelligence by positioning data as a strategic business asset [24].

Key characteristics of data monetization:

- **Direct Revenue Generation**: Selling data or data-derived products, such as insights or analytics, to third parties for profit. This can include raw data, aggregated reports, or predictive models tailored to specific industries [23].
- **Value-Added Services**: Leveraging data to improve existing products or create new services, such as personalized recommendations or targeted advertising, to improve customer engagement and generate additional revenue streams [22].

In this project, external data monetization through a decentralized Insights-as-a-Service (IaaS) model is focused. Instead of providing raw data, our system enables users to contribute verified and anonymized data points, which are then transformed into actionable insights. These insights primarily support specific applications in industry, offering institutions a more accurate and inclusive assessment for decision-making. Data contributors are rewarded through utility tokens, allowing them to treat their data as an economic asset, and essentially, this addresses enterprise demand for data-driven intelligence that is derived from better quality, owner-contributed, and less polluted data with verifiable provenance.

6. Personal Data as a Digital Asset: The Concept of Personal Data Assetization

Personal data assetization, or the conceptualization of personal data as a digital asset, denotes the socio-technical and legal processes through which personal information is transformed into a recognized digital asset [25], [26]. This involves structuring and treating personal information as a legitimate digital asset, rendering it capable of ownership, regulatory compliance, and participation in economic exchange.

The assetization process relies on three interrelated mechanisms: technical architecture, legal frameworks, and economic methods [25]-[28]. In this project, these mechanisms are demonstrated. Technically, algorithms for data verification and structuring in decentralized wallets transform raw travel data into verifiable blockchain-based tokens representing ownership. Legally, consent frameworks and smart contracts ensure user uphold their control over data access, analogous to property rights, and prevent unauthorized exploitation. Economically, these tokens enable monetization by assigning tangible value to the data, allowing users to trade, stake, or redeem them. Thus, overall, these effectively convert personal data into a tradable digital asset.

Comprehending these mechanisms, this turns people and their personal digital data into commodifiable metrics and sources of economic value, similar to other asset classes [4], [26]. This process closely parallels the tokenization of data into digital assets in our project.

These mechanisms shape the defining characteristics of personal data as a digital asset, particularly in decentralized architectures. The most relevant features for this project are summarized below.

Key characteristics of Personal Data Assetization:

- **Value Creation**: Personal data generates value in two key ways: utility, by improving services for individuals, and tradability, by enabling controlled economic exchange [26].

In this project, the value of data stems from its utility in generating personalized insights for alternative credit scoring, such as inferring financial stability from consistent travel spending patterns, whereby users are compensated via tokens for granting businesses access to insights through smart contracts. These tokens can subsequently be staked to earn rewards or redeemed for discounts.

- **Identifiability**: Personal data is typically digitally identifiable and linked to an individual, either directly, such as name or email, or indirectly, using a code or pseudonym [26].

In decentralized architectures, it is important to associate data with the correct individual while preserving data contributors' privacy to facilitate the process of assetizing personal data.

1.5 Project Scope

By design, a decentralized data wallet (DDW) ideally should be versatile, supporting diverse personal data types, ranging from behavioural and financial to health and educational, as well as supporting multiple data modalities [14]. It should also serve a variety of sector-specific applications and thus should not be confined to a single specific data domain, data modality, or industry-applicable use case. To ensure the implementation feasibility and maintain a focused evaluation of the DDW framework, this project deliberately narrows its focus to a single type of personal data, one application scenario, and four selected data modalities, which will serve as the basis for the proof-of-concept implementation of the DDW.

1.5.1 Focused Data Domain: Travel-Related Behavioural Data

To ensure the feasibility of implementation and maintain a focused evaluation of the decentralized data wallet (DDW) framework, this project deliberately narrows its scope to a single domain of personal data.

Following our initial round of literature review, travel-related behavioral data is selected as the proof-of-concept dataset for our DDW. This choice reduces the complexity of handling multiple heterogeneous data types while providing a clear and well-defined dataset for testing the architecture. Representative travel-related behavioural data includes airline choices, hotel stays, and overseas dining records as examples. Travel data is particularly suitable due to its practicality for collection, compliance with existing data privacy regulations, and carrying strong analytical value for downstream financial applications.

By concentrating on this specific data domain, the project is able to demonstrate the viability of the DDW architecture while keeping the evaluation process methodologically clear, manageable, and replicable for future extensions into broader categories of personal data.

1.5.2 Specific Use Case: Alternative Credit Scoring

A decentralized data wallet (DDW) should encompass and support a wide range of applications, spanning financial inclusion, human resources, healthcare, and more.

In our project, it narrows its focus to alternative credit scoring as the primary application. This use case provides a concrete context for generating actionable insights from travel-related behavioral data.

Alternative credit scoring refers to the use of non-traditional data sources to assess an individual's creditworthiness [29]. It enables lenders to evaluate the credit risk and repayment capacity of certain populations or individuals that may often be overlooked by mainstream credit systems, such as young adults with limited or no formal credit history, gig economy workers, or individuals in emerging markets. Thus, travel-related behavioural data is explored as a supplementary input for credit risk assessment.

Research by CGAP [30] demonstrates that transactional data, such as payment histories, can effectively assess credit risk for micro- and small-enterprises in emerging markets, supporting the broader applicability of non-traditional data sources in alternative credit scoring. This aligns with our project's exploration of travel-related behavioral data as a supplementary input for evaluating creditworthiness. It indicates that patterns such as frequent travel destinations, transportation choices, and overall mobility behaviour provide valuable insights into an individual's financial reliability [29],[30]. By integrating these specific behavioural insights, this can enhance the accuracy of credit risk assessment and broaden the inclusivity of lending decisions by credit scoring agencies and lenders.

By aligning with these insights, the project demonstrates how DDWs can bridge the gap between innovative data governance and real-world financial inclusion.

Ultimately, the proof-of-concept illustrates how financial institutions, such as banks, virtual lenders, and alternative scoring agencies, can make more informed, data-driven decisions when empowered with decentralized and user-consented data. Thus, not only does this enhance credit scoring effectiveness, but it also safeguards individual privacy and autonomy through a token-based, consent-driven decentralized data wallet.

1.5.3 Selected Data Modalities and Their Practicality

Finally, data contributors should allow to contribute and upload multiple data modalities (multimodal data), which includes, text (textual data), images & video (visual data), voice and sound track (audio data), digital footprint (behavioural data) and more, to create comprehensive personal profiles for more accurate insight generations for data consumers [14],[31]. Such comprehensiveness and flexibility are critical for unlocking the full potential of personal data as a secure and user-governed digital asset in the emerging data economy.

However, considering both the direct relevance of each data modality for generating accurate insights for alternative credit scoring and the time constraints of developing a proof-of-concept as a deliverable. (1) Structured text data, (2) Time-series data, (3) Image, (4) PDF documents are selected as the four primary modalities for our decentralized data wallet due to their direct relevance to generating accurate insights for alternative credit scoring and practicality in implementation [31],[32].

The project demonstrates a user-centric system in which data contributors securely upload travel data to the wallet, and data consumers, such as lenders, can access verified and anonymized insights for alternative credit scoring with user consent via smart contracts. This approach ensures secure, decentralized interactions while maintaining individual control over personal data [33].

Future work envisions scaling the wallet architecture to accommodate additional data domains with multiple data modalities. Thus, this expansion would enable the generation of insights for a variety of commercial applications, supporting a comprehensive, user-controlled data economy that prioritizes privacy, consent, and equitable value sharing.

To conclude, the project's focus and narrowed scope are summarized as follows:

- **Data Domain**: Travel-related behavioural data
- **Data Modalities**: Structured text, images, PDF documents, time-series data.
- **Use Case**: Alternative credit scoring for financial inclusion.
- **Data Contributors (Data Owners)**: Individuals (e.g., travelers, young adults, unbanked) who upload travel data to the wallet.
- **Data Consumers**: Financial institutions (traditional/virtual banks, lenders, credit scoring agencies) accessing anonymized insights for credit decisions.

1.6 Project Deliverables

Our project deliverables in this project are organised into three comprehensive and interdependent components, with each contributing to the overarching goal of empowering individuals with secure, actionable control over their travel data and demonstrating its real-world applicability for alternative credit scoring.

1.6.1 Technical Protocol Specification

This deliverable will be a comprehensive, documented specification on the technical protocols of the decentralised data wallet (DDW). It will provide clear and rigorous explanations on how all the architectural and functional layers are defined and logically integrated in the wallet, in accordance with established practices in blockchain, peer-to-peer data sharing, and tokenization. In this part, we focus on ensuring consistency between the technical layers and the system's logical architecture.

The specification will include:

- A systematic description of the operational flow of the decentralised data wallet, blueprinting the end-to-end process from user registration and data upload through tokenization, consent management, insight generation, and user compensation via tokenomics.
- An illustration of the overall architecture of the decentralised data wallet, providing a structural overview of modular components, from frontend user wallet interfaces to backend services and decentralized elements.

Building on this, the document will decompose the overall architecture into a detailed layered structure to explain its respective layered technical protocols. The layers include:

- Blockchain layer
- Smart Contract and Tokenization Layer
- Data Verification layer
- Privacy-Preserving Layer
- Decentralized Data Storage Layer
- User Consent and Access Control Layer

- Insight Generation and Delivery Layer
- Tokenomics Compensation and Reward Incentive Layer
- Security and Auditing Layer

Our ultimate goal is to formulate a robust and feasible technical protocol for our decentralised data wallet, offering a documented and referenceable protocol standard that comprehensively demonstrates the interactions between data contributors and data consumers. This ensures our decentralised data wallet is both theoretically sound and practically implementable while promoting innovation in P2P data economies.

1.6.2 Technical Architecture Design and Implementation

Building on the technical protocol specifications from 1.6.1, and proceeding to the second part of our deliverable, we will by then clearly understand how each layer of the technical architecture in the decentralized data wallet functions and how they can integrate together to provide a seamless user experience for both data contributors and data consumers.

1.6.2.1 Design and Implementation: System Architecture, Backend, and User Interface

Our second main deliverable is a web-based decentralized data wallet. It will focus on the design and practical implementation of the functioning of the decentralized data wallet through a proof-of-concept (PoC) demonstration. This PoC will embody all architectural and protocol specifications identified in the previous stage, demonstrating how the decentralised data wallet is implemented within a user-centric data ecosystem.

The prototype will showcase the following key features:

- **Data Verification via Oracles**: Uploaded data is validated by a decentralised oracle network for authenticity before tokenization.
- **Data Tokenization**: Convert travel-related behavioural data into ownership tokens, representing user ownership and enabling monetization in the wallet
- **Peer-to-Peer (P2P) Data Sharing Mechanism**: Data consumers can request access to specific data contributors' data profiles via smart concepts. Data contributors can grant

time-bound consent for data consumers to access specific insights for alternative credit scoring.

- **Privacy-Preserving Insight Generation:** Anonymized credit scoring insights are generated using rule-based or machine learning models, protected by zero-knowledge proofs to safeguard users' privacy.
- **Automated Tokenomics Compensation:** Data contributors receive utility tokens for sharing insights. These tokens can be staked or redeemed for rewards. In turn, businesses gain actionable credit insights, supporting a sustainable and transparent token economy.

1.6.2.2 Proof-of-Concept Validation through Beta Testing and User Feedback Collection

Followed by the technical architecture development of the decentralised data wallet, this proof-of-concept will then undergo a beta testing phase with approximately 20 participants acting as data contributors. This sub-deliverable will focus on validating the PoC of the decentralized data wallet through users' participation.

In this part, we will systematically capture the workflow interactions between data contributors and data consumers in order to confirm the wallet's functionality, usability, and effectiveness in a user-centric data ecosystem.

Participants will engage with the digital wallet system to perform tasks including uploading data, verifying data ownership through tokenization, managing their data profiles, granting consent for profile access, and collecting rewards in the form of utility tokens. Participants' feedback in quantitative metrics and qualitative insights will be collected to evaluate our artifact technically and operationally. Subsequently, the results will be used for iterative refinements to the wallet's architecture and user interface, ensuring it meets both technical and user-centred requirements before broader real-world applications.

The output of the evaluation will be a comprehensive validation report, documenting the following elements based on post-beta testing data:

- **Testing Outcomes:** Quantitative metrics (detailed in Section 2.2.2) demonstrating that the wallet performs as intended across core functionalities.

- **User Feedback:** Synthesized qualitative insights from participants, capturing perceptions of usability, clarity, trust, and overall experience for design considerations and improvement.
- **Refinement Actions:** Targeted updates to the wallet's features, interface, or underlying architecture based on collected quantitative and qualitative feedback.

This validation summary will demonstrate that the PoC of the decentralised data wallet meets predefined technical and operational criteria, ensuring it is robust and user-centric for alternative credit scoring applications.

1.6.3 Discussion on the Practicality and Limitations in Alternative Credit Scoring Applications

The final deliverable will present an evaluative discussion on the practicality and limitations of the decentralized data wallet (DDW) in supporting alternative credit scoring through the use of travel-related behavioral data.

This discussion summary is structured to present and synthesize findings from three complementary perspectives to provide a balanced assessment of the real-world applicability of DDWs in alternative credit scoring. These perspectives include: (1) **quantitative metrics to assess DDW's system performance** (specific metrics include insight generation effectiveness and tokenomics compensation efficiency, as detailed in Section 2.3) to evaluate system feasibility; (2) **feedback from beta testing participants** to highlight user-centric challenges; and (3) **insights from industry practitioners** in fintech or traditional finance, to address real-world integration and regulatory constraints.

Each perspective contributes a distinct and complementary view on DDW's application for alternative credit scoring. By combining the unique insights provided by each perspective, this discussion provides a comprehensive understanding of the DDW's operational feasibility, scalability constraints, regulatory compliance, and user adoption challenges in real-world financial and regulatory contexts.

2. Project Methodologies

To achieve the project objectives and develop the project deliverables, we divide our project into four stages. In each stage, each methodology employed will correspond to the respective deliverable as outlined in Section 1.6.

2.1 Multivocal Literature Reviews

To produce the detailed specification of technical protocols for the DDW in Section 1.5.1, a Multivocal Literature Review (MLR) will be adopted.

Multivocal Literature Reviews (MLR) is a systematic research method that integrates both formal, published academic literature (scholarly research) and grey literature (practitioner sources) [34]. Examples of grey literature include industry white papers, technical reports, standards documentation, conference proceedings, companies' case studies, and in-field practitioner blogs. Unlike conventional Systematic Literature Reviews (SLR) that rely exclusively on peer-reviewed academic literature, MLR is particularly valuable in emerging and fast-evolving research domains in the field of software engineering, as insights from industry practitioners are essential for understanding how research translates into real-world applications [34], [35].

Given the novelty of decentralised data wallets and rapidly evolving paradigms of blockchain protocols, decentralised finance and decentralised platforms for data sharing, academic coverage in relevant scopes is limited and often lags behind industrial practice, while industries that focus on enterprise blockchain solutions and decentralized data storage networks often advance technical protocols and implementation frameworks more rapidly than academia [35].

MLR allows us to deliberately acknowledge and synthesize insights from multiple perspectives from both the research community and industry community. By employing MLR, we aim to provide a comprehensive overview of a research area's state-of-the-art and state-of-practice in industry. Thus, this ensures our technical protocol specification is both theoretically grounded and practically implementable.

To employ MLR effectively in developing the technical protocols for our decentralised data wallet, we will first define the review questions to guide our scope of inquiry. In regard to our project topic, we have set the following review questions as search parameters to guide our literature reviews towards developing the technical protocols of the decentralised data wallet:

No.	Review Questions
#1	What are the current and/or proposed technical protocols and architectural designs for decentralized data wallets, particularly for managing personal or financial data in blockchain environments?
#2	What protocols support secure data storage, tokenization, and ownership representation for data assets, such as travel-related data points as utility tokens?
#3	What are effective practices for data verification using oracles or similar mechanisms to ensure authenticity in decentralized systems?
#4	How do smart contracts facilitate user consent with time-bound access and token transactions for data sharing and user compensation ?
#5	What and how privacy-preserving techniques, such as zero-knowledge proofs, anonymization, and encryption methods, are used when generating anonymized insights from shared data?
#6	What insights can be derived from non-traditional data, or in particularly travel-related behavioural data, to improve the effectiveness of alternative credit scoring?
#7	What rule-based methods or simple machine learning methods could be used to generate personalized credit scoring insights from travel-related behavioural data?

Next, relevant sources will be collected by searching academic databases and grey literature sources from industrial practitioners in decentralised finance and companies' whitepapers and technical reports. To ensure relevance to the emerging field of decentralization applied to personal data ownership and management, recent sources published between 2020 and 2025 will be prioritised, capturing the latest advancements in blockchain-based data sharing and decentralised data management mechanisms, and extending their applications in travel data monetization and alternative credit scoring. Finally, actionable insights are extracted from selected sources. Relevant insights will be used to synthesize a cohesive technical protocol specification document for our decentralised data wallet.

2.2 Technical Architecture Design, Implementation and Testing

2.2.1 Proof-of-Concept (PoC) Development

The methodology of Proof-of-Concept (PoC) development is employed for the design and implementation of the decentralized data wallet. In this stage, we will develop the technical protocols into a minimal but functional system that demonstrates the feasibility of our proposed artifact and validate its core functionalities and workflow.

This PoC delivers a web-based decentralized data wallet. The major components of the decentralized data wallet prototype would primarily include: (i) Frontend Interface, (ii) Backend logic, (iii) Blockchain Layer, (iv) Smart Contract and Tokenization Implementations, (v) Decentralised Data Storage, (vi) Mock Decentralised Oracle Network Service for Data Verifications, (vii) Privacy-Preserving Layer and the (viii) Insight Generation module and delivery. These components collectively enable the end-to-end transaction flow, from data upload to insight delivery to data consumers and token compensation to data contributors.
The Ethereum testnet is selected as the core blockchain layer platform for our decentralized data wallet [36]. Ethereum testnet is a simulated blockchain environment that mirrors Ethereum's mainnet functionality, using valueless test Ether for free development and testing of smart contracts and tokenization [36]. It is chosen for our project because it provides a cost-effective and robust environment for developing and testing smart contracts and tokenization for our architect at an initial stage.

2.2.2 Internal Technical and Operational Evaluation

After developing our prototype and before launching beta testing to involve end-users, the PoC undergoes a rigorous technical and operational assessment to confirm that the system functions correctly and efficiently.

From a **technical perspective**, the evaluation will verify:

- **Data-to-Asset Transformation**: whether travel-related data is securely converted into Data ownership tokens.
- **Smart Contract Functionality**: whether smart contracts execute correctly, handling data access requests, consent, time-bound permissions, and token-based compensation.
- **Data Privacy and Security** – whether privacy-preserving techniques and access controls maintain data integrity and prevent unauthorized access.

From an **operational perspective**, the evaluation will address:

- **Layered Architecture Integrity**: whether the different architectural components operate cohesively across the system.
- **System Performance and Reliability**: how efficiently the wallet performs under real-time usage conditions.

This internal evaluation ensures the PoC's technical robustness and operational effectiveness and prevents user frustration or unreliable feedback during subsequent beta testing.

2.2.3 Beta testing and User-Centered Design (UCD) Evaluation

After the internal technical and operational evaluation of the PoC, we will proceed to the testing stage. In this stage, we will employ Beta Testing with a User-Centred Design (UCD) evaluation as our testing methodology.

Besting testing allows us to verify and check whether the decentralised data wallet PoC is validated not only in terms of technical functionality, but also whether the system works for actual users. On the contrary, UCD evaluation ensures the system is designed around users'

needs, experiences, and expectations. Together, these provide a comprehensive testing process to validate the PoC holistically across functionality, usability, and trust dimensions.
For beta testing, we will invite 20 participants to test the prototype as data contributors by interacting with the decentralized data wallet prototype. These participants will be asked to perform representative tasks, including wallet registration, uploading travel receipts and any relevant travel-related behavioural data, responding to business data access requests, and receiving compensation through TRVL utility tokens. Complementing this, a smaller set of participants representing potential alternative credit scoring agencies or lenders, as data consumers, interacted with the system's dashboard to request and interpret insights. Thus, this simulates both sides of the ecosystem and provides realistic validation of how the wallet operates within a two-sided market.

Following beta testing, participant feedback will be collected using a mixed-methods approach, with both quantitative and qualitative insights collected. The latter will be our emphasis on evaluation due to the novelty of our architecture. Table 1 shows the qualitative measures (numbers) and qualitative insights (opinions and perceptions) that we initially established as the basis for evaluating the PoC:

Type of Data	Measurement & Method
Quantitative measures	• Task completion rate (how many finished tasks successfully) • Error rate (how many mistakes occurred) • Usability scores (System Usability Scale = a standard 10-question survey for usability)
Qualitative insights	• Semi-structured interviews (guided conversations to know the user experience) • Think-aloud protocol (users verbalize what they think while using the system)

Table 1. Evaluation Measures and Methods for Decentralized Data Wallet Prototype

This dual strategy ensures that the UCD evaluation captures both measurable performance outcomes and users' perceptions of trust and overall ease of use, which are factors essential for adoption in fintech contexts.

Participant feedback will be collected and analyzed through descriptive statistical methods for the quantitative metrics and thematic coding for qualitative feedback. The results will be compiled into a list of the main usability issues and suggested design improvements for refinements. Consequently, this iterative feedback loop ensures our PoC could be adjusted to align with users' expectations and practical requirements, thereby further enhancing the wallet's overall acceptability and functionality in real-world contexts.

2.3 Limitations & Practicality Empirical Evaluation with Mixed-Methods Analysis

To discuss the practicality and limitations of the decentralized data wallet (DDW) for alternative credit scoring, the methodology of Empirical Evaluation with Mixed-Methods Analysis will be used. Mixed-Methods Analysis combines numerical data with observational or narrative data to gain a more complete understanding of a system's effectiveness and real-world industrial applicability. By capturing objective system metrics, user experience, and industry feasibility, this method is critical for assessing practicality and limitations in alternative credit scoring applications in real-life contexts.

As mentioned in Section 1.6.3, three complementary perspectives are considered for evaluating the practicality and limitations of the decentralized data wallet (DDW) for alternative credit scoring: (1) quantitative metrics assessing DDW's system performance, (2) feedback from beta testing participants, and (3) insights from industry practitioners. Below, each perspective is described in detail with its purpose and discussion points focused.

1. Quantitative Metrics

Quantitative metrics objectively measure how well the DDW handles core processes required for alternative credit scoring, thereby providing measurable evidence of technical limitations that may affect real-world usability and operational feasibility.

Discussion Points:

- **Insight Generation Effectiveness:** measures how accurately the DDW derives credit-relevant "insights" from tokenized data. High failure rates or low precision in insight generations, such as false positives, indicate potential limitations in using non-traditional data, such as travel-related behavioral data, for better decisions in alternative credit scoring.
- **Tokenomics Compensation Efficiency**: measures how accurately the DDW distributes rewards (tokens) to users for sharing data, a key incentive in P2P economies, ensuring fair compensation motivates participation in credit scoring ecosystems. Low efficiency or errors in token distribution can limit the wallet's practicality by reducing user trust and discouraging engagement, thereby affecting the scalability and real-world adoption of the system.

2. Feedback from Beta Testing Participants

Feedback from end-users evaluates the trustworthiness of individuals, as data contributors, to the DDW, which is critical for adoption in alternative credit scoring systems.

Discussion Points:

- **Level of Perceived Privacy:** Participants' perceptions of data security and privacy provide insights into how much they trust the DDW. Low confidence in data protection or unclear privacy controls signals will lessen individuals' willingness to monetize personal data, let alone for alternative credit scoring purposes.
- **Level of Reward Satisfaction:** Participants' satisfaction with TRVL token compensation provides insight into their motivation to engage with the system. If they are dissatisfied with the reward mechanism, it is hard to broaden the user base of the DDW to make individuals get engaged in the data monetization ecosystem.

3. Insights from Industry Practitioners

Insights from current industry practitioners, including credit analysts, alternative lending officers, fintech product specialists, and data privacy compliance specialists, guide the evaluation to address real-world applicability, compliance, and integration feasibility, which are crucial for practical implementation in the industrial landscape.

Discussion Points:

- **Feasibility of Integration into Financial Decision-Making:** Practitioners assess whether DDW-generated insights can be meaningfully incorporated into existing alternative credit scoring workflows for real financial decision-making.

- **Regulatory Constraints:** Certain regulations may restrict the widespread adoption of the DDW for alternative credit scoring. Examples are ***data privacy laws*** that govern how personal data can be stored and shared, ***financial regulations*** that set standards for credit risk and open banking, and ***data localization laws*** that require personal data to remain within national boundaries. These constraints could limit the widespread use of DDWs globally, as non-compliance may result in legal and operational risks.

- **Scalability Limitations:** Scalability refers to the ability of the DDW to handle growing numbers of users, data uploads, and transactions without performance degradation. Limitations may arise from ***blockchain throughput (transaction per second bottlenecks)***, ***high gas fees*** when moving beyond testnets, ***oracle request delays*** during data verification, or rising storage needs as more travel data is added. These can hinder the practicality of deploying DDWs in real-world credit scoring, where systems must support large and diverse user bases efficiently.

To summarise, Table 2 outlines the key perspectives, purposes, and discussion points used to evaluate the practicality and limitations of DDWs in alternative credit scoring.

Key Perspectives	Purposes	Discussion Points
Quantitative Metrics	Evaluate technical limitations towards real-world usability and operational feasibility	• Insight Generation Effectiveness • Tokenomics Compensation Efficiency
Feedbacks from Beta Testing Participants	Evaluate data contributors' trustworthiness to DDW	• Level of Perceived Privacy • Level of Reward Satisfaction
Insights from Industry Practitioners	Evaluate real-world adoption feasibility in financial contexts.	• Feasibility of Integration into Financial Decision-Making • Regulatory Constraints • Scalability Limitations

Table 2. Discussion Points for Evaluating the Limitations and Practicality of DDWs in Alternative Credit Scoring Applications

3. Project Schedule and Milestones

<table>
<tr><th>Time Period</th><th>Tasks</th><th>Completion</th></tr>
<tr><td>September</td><td>Meet with supervisor</td><td>Done</td></tr>
<tr><td></td><td>Phase 1 Deliverable
Detailed project plan
Project website</td><td>Done</td></tr>
<tr><td></td><td>Do literature reviews on tokenization and tokenomics in P2P data ecosystems, blockchain protocols and architecture, privacy-preserving techniques, and alternative credit scoring models
Identify suitable behavioural data-specific domain, modalities, and their derived insights</td><td>Done</td></tr>
<tr><td>October</td><td>Case studies review:
Decentralised P2P data marketplace
Self-sovereign identity specifications
Decentralised AI machine training
Refine review questions based on MLR findings
Develop and draft technical protocol
Define data types, token standards, access control logic and layered architecture</td><td></td></tr>
<tr><td>November</td><td>PoC Development (Phase 1): frontend interface</td><td></td></tr>
<tr><td>December</td><td>PoC Development (Phase 2): system architecture
Backend logic
Blockchain integration
Smart contract development</td><td></td></tr>
<tr><td>January</td><td>PoC Development (Phase 3): integration
Connect backend with frontend
Decentralised oracle networks
Prepare for first presentation</td><td></td></tr>
<tr><td></td><td>Phase 2 Deliverable
Interim Report</td><td></td></tr>
<tr><td>February</td><td>Conduct internal technical & operation evaluation
Conduct beta testing
Collect beta testing feedbacks
PoC Development (Phase 4): refinement after beta testing
Evaluation on practicality & limitations (Phase 1):
Analyze quantitate metrics
Analyze beta testing feedbacks</td><td></td></tr>
<tr><td>March-April</td><td>Evaluation on practicality & limitations (Phase 2):
Get insights from industry practitioners
Write final report
Prepare for final presentation
Finalise project website with PoC demo</td><td></td></tr>
<tr><td>April</td><td>Phase 3 Deliverable
Final report</td><td></td></tr>
</table>

4. References

[1] [1] P. Hummel, M. Braun, M. Tretter, and P. Dabrock, "Data sovereignty: A review," *Big Data & Society*, vol. 8, no. 1, pp. 1–17, Jan. 2021. [Online]. Available: https://pure.tue.nl/ws/portalfiles/portal/188041522/2053951720982012.pdf. [Accessed Sep. 10, 2025].

[2] R. Baraku et al., "Personal Data Sovereignty through Federated Access and Policy Control," presented at the PhD Workshop at VLDB 2025, virtual, Aug. 2025. [Online]. Available: https://www.vldb.org/2025/Workshops/VLDB-Workshops-2025/PhD/PhD25_14.pdf. [Accessed Sep. 10, 2025].

[3] K. Birch, D. T. Cochrane, and C. Ward, "Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech," *Big Data & Society*, vol. 8, no. 2, pp. 1–15, Jul.–Dec. 2021, doi: 10.1177/20539517211017308. [Online]. Available: https://journals.sagepub.com/doi/10.1177/20539517211017308. [Accessed Sep. 11, 2025].

[4] World Economic Forum, "Personal Data: The Emergence of a New Asset Class," Geneva, Switzerland, Feb. 2011. [Online]. Available: https://media.bain.com/Images/WEF_Personal_Data_A_New_Asset_Class_Telecom_opportunities.pdf. [Accessed Sep. 12, 2025].

[5] A. Acquisti, C. Taylor, and L. Wagman, "The Economics of Privacy," *Journal of Economic Literature*, vol. 54, no. 2, pp. 442–492, Jun. 2016, doi: 10.1257/jel.54.2.442. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2580411. [Accessed Sep. 12, 2025].

[6] Pew Research Center, "Americans and Privacy: Concerned, Confused and Feeling Lack of Control Over Their Personal Information," Washington, DC, USA, Nov. 15, 2019. [Online]. Available: https://www.pewresearch.org/internet/2019/11/15/americans-and-privacy-concerned-confused-and-feeling-lack-of-control-over-their-personal-information/. [Accessed Sep. 14, 2025].

[7] K. Nabben, "Value from Data? A Decentralized Approach," *Information and Organization*, vol. 35, no. 3, pp. 1–15, Jul. 2025, doi: 10.1177/29768624251358645. [Online]. Available: https://journals.sagepub.com/doi/10.1177/29768624251358645. [Accessed Sep. 14, 2025].

[8] N. E. Obeidat et al., "Identifying and Addressing Data Asymmetries to Enable Better Data Sharing," *JAMIA Open*, vol. 5, no. 2, pp. 1–10, Jul. 2022, doi: 10.1093/jamiaopen/ooac054. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9339620/. [Accessed Sep. 14, 2025].

[9] Javelin Strategy & Research, "Identity as a digital asset: Tokenizing the self," 2024. [Online]. Available: https://javelinstrategy.com/research/identity-digital-asset-tokenizing-self. [Accessed Sep. 14, 2025].

[10] "The 10 data marketplaces platform revolutionizing data sharing and monetization," Dec. 16, 2024. [Online]. Available: https://scikiq.com/blog/the-10-data-marketplaces-platform-revolutionizing-data-sharing-monetization/. [Accessed Sep. 14, 2025].

[11] McKinsey & Company, "What is tokenization?" Jul. 24, 2024. [Online]. Available: https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-tokenization. [Accessed Sep. 14, 2025].

[12] M. M. Merlec and H. P. In, "DataMesh+: A blockchain-powered peer-to-peer data exchange model for self-sovereign data marketplaces," *Sensors*, vol. 24, no. 6, p. 1896, Mar. 2024, doi: 10.3390/s24061896. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC10974787/. [Accessed Sep. 15, 2025].

[13] F. D. Nobre et al., "A decentralized architecture for trusted dataset sharing using blockchain," *Sensors*, vol. 22, no. 23, p. 9535, Dec. 2022. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9736492/. [Accessed Sep. 16, 2025].

[14] A. Korir et al., "An empirical study of a decentralized identity wallet," presented at the *USENIX Symposium on Usable Privacy and Security*, virtual, Aug. 2022. [Online]. Available: https://www.usenix.org/system/files/soups2022-korir.pdf. [Accessed Sep. 16, 2025].

[15] ETSI, "Zero-knowledge proofs and selective disclosure," ETSI TR 119 476 V1.1.1, European Telecommunications Standards Institute, Sophia Antipolis, France, Apr. 2023. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/119400_119499/119476/01.01.01_60/tr_119476v010101p.pdf. [Accessed Sep. 16, 2025].

[16] Dock Labs, "Self-sovereign identity: The ultimate guide 2025," Dec. 2023. [Online]. Available: https://www.dock.io/post/self-sovereign-identity. [Accessed Sep. 16, 2025].

[17] K. Zheng et al., "A comprehensive review of blockchain technology: Underlying mechanisms and applications," Blockchain: Research and Applications, vol. 4, no. 2, p. 100184, Dec. 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772662223001844. [Accessed Sep. 17, 2025].

[18] D. R. Kuhn, D. Yaga, and J. Voas, "Blockchain technology overview," NISTIR 8202, National Institute of Standards and Technology, Gaithersburg, MD, USA, Oct. 2018. [Online]. Available: https://nvlpubs.nist.gov/nistpubs/ir/2018/nist.ir.8202.pdf. [Accessed Sep. 17, 2025].

[19] V. Baraku, E. Ramadani, I. Paraskakis, S. Veloudis, and P. Yadav, "Defining personal data sovereignty: An ontologically-based framework facilitating subject privacy control," Data and Information Management, vol. 9, no. 1, p. 100108, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2543925125000166. [Accessed Sep. 18, 2025].

[20] M. S. Mayernik et al., "Data Conservancy Provenance, Context, and Lineage Services: Key Components for Data Preservation and Curation," Data Science Journal, vol. 12,

pp. 158–171, 2013. [Online]. Available: http://doi.org/10.2481/dsj.12-039. [Accessed Sep. 19, 2025].

[21] S. D. Viglas, "Data Provenance and Trust," *Data Sci. J.*, vol. 12, pp. GRDI58–GRDI63, 2013. [Online]. Available: https://www.researchgate.net/publication/270166696_Data_Provenance_and_Trust. [Accessed: Sep. 22, 2025].

[22] B. H. Wixom, C. M. Beath, and L. Owens, "What is Data Monetization?," *MIT CISR Res. Briefing*, Aug. 2023. [Online]. Available: https://cisr.mit.edu/publication/2023_0801_DataMonetization_WixomBeathOwens. [Accessed: Sep. 22, 2025].

[23] J. Ofulue and M. Benyoucef, "Data monetization: insights from a technology-enabled literature review and research agenda," *Manag. Rev. Q.*, vol. 74, no. 2, pp. 535–580, 2024. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9707275/. [Accessed: Sep. 23, 2025].

[24] Sisense, "Insights as a Service (IaaS)," Jul. 21, 2025. [Online]. Available: https://www.sisense.com/glossary/insights-as-a-service-iaas/. [Accessed: Sep. 23, 2025].

[25] T. Beauvisage and K. Mellet, "Datassets: Assetizing and Marketizing Personal Data," in *Assetization: Turning Things into Assets in Technoscientific Capitalism*, K. Birch and F. Muniesa, Eds. Cambridge, MA, USA: MIT Press, 2020, pp. 75–95. [Online]. Available: https://direct.mit.edu/books/oa-edited-volume/4848/chapter/625226/Datassets-Assetizing-and-Marketizing-Personal-Data. [Accessed: Sep. 23, 2025].

[26] K. Birch, D. T. Cochrane, and C. Ward, "Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech," *Big Data Soc.*, vol. 8, no. 1, Jan.–Jun. 2021. [Online]. Available: https://eprints.whiterose.ac.uk/id/eprint/217437/. [Accessed: Sep. 24, 2025].

[27] OECD, *Exploring the Economics of Personal Data: A Survey of Methodologies for Measuring Monetary Value*, OECD Digital Economy Papers, no. 220. Paris, France: OECD Publishing, 2013. [Online]. Available: https://www.oecd.org/content/dam/oecd/en/publications/reports/2013/04/exploring-the-economics-of-personal-data_g17a228d/5k486qtxldmq-en.pdf. [Accessed: Sep. 25, 2025].

[28] F. Ma, Y. Sun, S. Xiong, and W. Wang, “An exploration of the data assetization paths for the three major types of data,” *J. Inf. Resour. Manag.*, vol. 14, no. 5, pp. 4–13, 2024. [Online]. Available: https://jirm.whu.edu.cn/jwk3/xxzyglxb/EN/10.13365/j.jirm.2024.05.004. [Accessed: Sep. 25, 2025].

[29] World Bank Group, “The use of alternative data in credit risk assessment: Opportunities, risks, and challenges,” World Bank, Washington, DC, USA, 2024. [Online]. Available:

https://documents1.worldbank.org/curated/en/099031325132018527/pdf/P179614-3e01b947-cbae-41e4-85dd-2905b6187932.pdf. [Accessed: Sep. 25, 2025].

[30] M. Karim *et al.*, “Leveraging transaction data and alternative behavioral signals for credit scoring,” *J. Financial Services Res.*, vol. 64, no. 1–2, pp. 45–67, Aug. 2023. [Online]. Available: https://www.cgap.org/research/publication/leveraging-transactional-data-for-micro-and-small-enterprise-lending. [Accessed: Sep. 25, 2025].

[31] A. I. Hashi, “Machine learning-driven fintech solutions for credit scoring and financial inclusion in the gig economy,” *J. Financial Innov.*, 2025. [Online]. Available: https://scholar9.com/publication/4cfde31befce59a2d305b48f00d39e93.pdf. [Accessed: Sep. 25, 2025].

[32] T. Aslam and A. Aslam, “Social-Credit+: AI-driven social media credit scoring platform,” *arXiv preprint arXiv:2506.12099*, 2025. [Online]. Available: https://arxiv.org/abs/2506.12099. [Accessed: Sep. 25, 2025].

[33] Q. Wang and Y. Liu, “Blockchain empowered dynamic access control for secure data sharing in collaborative emergency management,” *Inf. Process. Manag.*, vol. 61, no. 3, Art. no. 103960, May 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306457324003194. [Accessed: Sep. 26, 2025].

[34] V. Garousi, M. Felderer, and M. V. Mäntylä, “Guidelines for including grey literature and conducting multivocal literature reviews in software engineering,” *Inf. Softw. Technol.*, vol. 106, pp. 101–121, Feb. 2019. [Online]. Available: https://pureadmin.qub.ac.uk/ws/files/178657399/GuidelinesforconductingMLRs_Sept18.pdf. [Accessed: Sep. 26, 2025].

[35] P. Soares, A. A. Araújo, G. Destefanis, R. Neykova, R. Saraiva, and J. Souza, “Blockchain developer experience: A multivocal literature review,” *arXiv preprint arXiv:2501.11431*, 2025. [Online]. Available: https://www.researchgate.net/publication/388231686_Blockchain_Developer_Experience_A_Multivocal_Literature_Review. [Accessed: Sep. 27, 2025].

[36] M. ElSheikh, D. Draheim, and K. van Overveld, “Blockchain-enabled decentralized identity management: The case of self-sovereign identity in public transportation,” *J. King Saud Univ. Comput. Inf. Sci.*, vol. 33, no. 8, pp. 944–952, Oct. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2096720921000099. [Accessed: Sep. 26, 2025].

[37] N. Naik and P. Jenkins, “Self-sovereign identity specifications: Govern your identity through your digital wallet using blockchain technology,” Defence School of Communications and Information Systems, Ministry of Defence, UK, and School of Computing, University of Portsmouth, UK, 2019. [Online]. Available: https://publications.aston.ac.uk/id/eprint/41998/1/SSI_Specifications_uPort_SovrinDrNitinNaik.pdf. [Accessed: Sep. 27, 2025].